

CSA5112 - Cryptography and Network Security

7. Columnar Transposition Cipher

Aim: To implement the Columnar Transposition Cipher for encrypting plaintext using a keyword.

Algorithm:

1. Start.
2. Read plaintext.
3. Arrange plaintext row-wise into columns.
4. Read columns in order according to the key.
5. Display ciphertext.
6. Stop.

Code:

```
#include<stdio.h>

#include<string.h>

int main()

{

    char text[]="HELLOWORLD";

    int cols=5;

    int rows=(strlen(text)+cols-1)/cols;

    int i,j,k=0;

    char mat[10][10];

    for(i=0;i<rows;i++)

    {

        for(j=0;j<cols;j++)

        {

            if(text[k]!='\0')
```

```
        mat[i][j]=text[k++];

    else

        mat[i][j]='X';

    }

}

printf("Cipher Text: ");

for(j=0;j<cols;j++)

    for(i=0;i<rows;i++)

        printf("%c",mat[i][j]);

return 0;

}
```

Sample Input:

```
Plain Text: HELLOWORLD
```

Sample Output:

```
Cipher Text: HWEOLRLLOD
```

Result:

The Columnar Transposition Cipher was successfully implemented, and the plaintext was encrypted by arranging characters into columns.